

KEY DEVELOPMENT

THE CONTEMPORARY ERA

The most significant developments in recent military technology have been driven not by increased firepower, but by the integration of electronics and computers into weapons systems; many modern conflicts bear witness to the advantage these technologies can provide.

The first steps towards precision-guided (“smart”) weapons date back to the final stages of World War II, but it was not until the late 20th century that these devices truly transformed the military environment. The combination of vast leaps in computer power with the miniaturization of electronics over recent decades has led to the development of projectiles with their own guidance systems, enabling exceptional accuracy over long ranges. This process has extended from the radar and heat-seeking guidance systems that are now commonplace in missiles, to some artillery shells, ground-launched rockets, and even man-portable projectile weapons.

Modern military aircraft are fitted with a huge array of electronics to improve missile accuracy and acquire targets alongside devices that help to counter anti-aircraft weapons. Remote-control guidance has been extended to the weapons themselves: the “drone”, or

Unmanned Aerial Vehicle (UAV), is one such example. This remote-controlled aircraft can fly over hostile territory to provide detailed photographic reconnaissance, or even conduct strike missions, without the risk of a pilot being killed or captured.

SOUND AND VISION

Battlefield communications have improved vastly over recent years. The “fog of war” – which prevented a general from knowing what was going on in battle – is mostly a thing of the past, with individual radios allowing officers and troops to communicate directly. Global Positioning Satellites (GPS) enable a commander back at base to maintain close control over patrols or armoured vehicles, with constant, real-time updates on their position and battlefield capability. For the troops on the frontline, laser range-finders improve the accuracy of weapons, and nightfall no

KEY ATTACK

AL-QAEDA ATTACKS AMERICA 11 SEPTEMBER 2001

The assault by al-Qaeda on the US homeland in 2001 exposed the inherent vulnerability of an open society to attack. The terrorists, armed only with knives, were able to hijack four civilian airliners, crashing two of them into the World Trade Center, and one into the Pentagon.



▲ The second aircraft crashes into the World Trade Center; 2,977 civilians died in the attacks that day.

▼ AFGHANISTAN

A US Army CH-47 Chinook helicopter provides aerial security in Khost Province, Afghanistan, in early 2012.

